Сетевые технологи

Лабораторная работа № 1. Методы кодирования и модуляция сигналов

Содержание

Список иллюстраций

1 Постановка задачи

Octave.

.

.

2 Выполнение лабораторной работы

2.1 Построение графиков в Octave

$y_1 = \sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x$ Octave.

. x y , .

`plot`, , 4

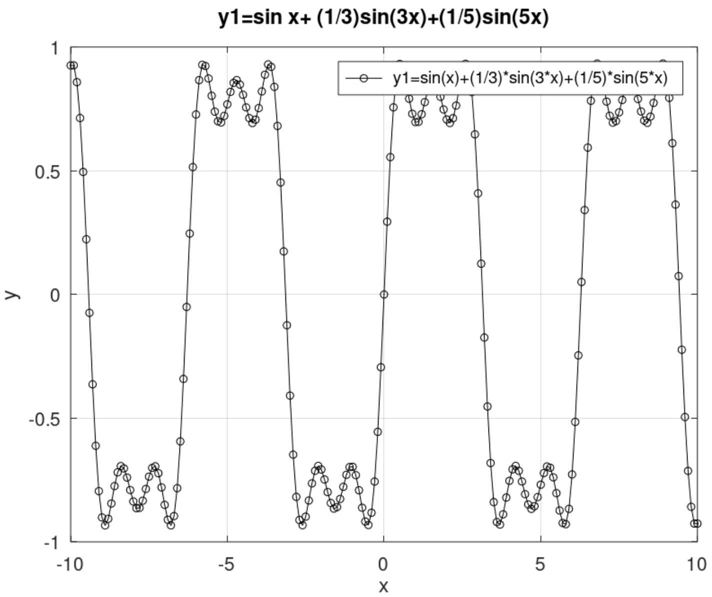
(1).

1. $y_1 = \sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x$

```
% x:
x = -10:0.1:10;
% y.
y1=sin(x)+1/3*sin(3*x)+1/5*sin(5*x);
% :
plot(x,y1, "-ok; y1=sin(x)+(1/3)*sin(3*x)+(1/5)*sin(5*x);","markersi
%
grid on;
% X:
xlabel('x');
% Y:
ylabel('y');
```

```
% :
title('y1=sin x+ (1/3)sin(3x)+(1/5)sin(5x)');
% .eps:
print ("plot-sin.eps", "-mono", "-FArial:16", "-deps")
% .png:
print("plot-sin.png");
```

(. **fig:001**)



2.1: $y_1 = \sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x$

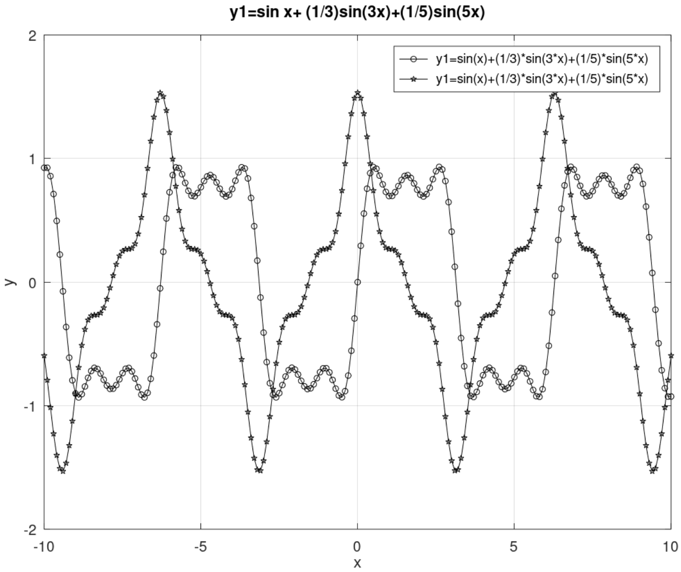
$y_1 = \sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x$ $y_2 = \cos x + \frac{1}{3}\cos 3x + \frac{1}{5}\cos 5x$. 1 `hold on` , (2).

2. $y_1 = \sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x$ $y_2 = \cos x + \frac{1}{3}\cos 3x + \frac{1}{5}\cos 5x$

```
% x:
x=-10:0.1:10;
% y.
y1=sin(x)+1/3*sin(3*x)+1/5*sin(5*x);
```

```
y2=cos(x)+(1/3)*cos(3*x)+(1/5)*cos(5*x);
% [illegible]:
plot(x,y1, "-ok; y1=sin(x)+(1/3)*sin(3*x)+(1/5)*sin(5*x);","markersi
% [illegible]
hold on;
% [illegible]:
plot(x,y2, "-pk; y2=cos(x)+(1/3)*cos(3*x)+(1/5)*cos(5*x);","markersi
% [illegible]
grid on;
% [illegible] X:
xlabel('x');
% [illegible] Y:
ylabel('y');
% [illegible] .eps:
print ("plot-sin1.eps", "-mono", "-FArial:16", "-deps")
% [illegible] .png:
print("plot-sin1.png");
```

(. **fig:002**).



2.2: $y_1 = \sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x \quad y_2 = \cos x + \frac{1}{3}\cos 3x + \frac{1}{5}\cos 5x$

2.2 Разложение импульсного сигнала в частичный ряд Фурье

m- , , .

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subplot plot .

plot (3).

3. , ,

```
% ☒☒☒☒☒☒☒☒☒☒ ☒☒☒☒☒☒☒☒ (☒☒☒☒☒☒☒☒):
N=8;
```

```
% [illegible] [illegible]:
t=-1:0.01:1;
% [illegible] [illegible]:
A=1;
% [illegible]:
T=1;
% [illegible] [illegible]
nh=(1:N)*2-1;
% [illegible] [illegible] [illegible] [illegible], [illegible] [illegible] cos:
Am=2/pi ./ nh;
Am(2:2:end) = -Am(2:2:end);
% [illegible] [illegible]:
harmonics=cos(2 * pi * nh' * t/T);
% [illegible] [illegible] [illegible]:
s1=harmonics.*repmat(Am',1,length(t));
% [illegible] [illegible]:
s2=cumsum(s1);
% [illegible] [illegible]:
for k=1:N
subplot(4,2,k)
plot(t, s2(k,:))
end
print("meandr_cos.png");
```

, 3 ,

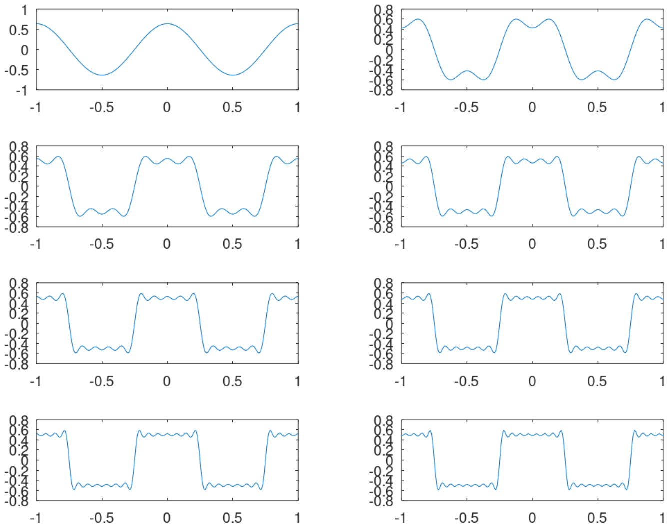
(4).

4. , ,

```
% [illegible] [illegible] ([illegible]):
N=8;
```

```
% [illegible] [illegible]:
t=-1:0.01:1;
% [illegible] [illegible]:
A=1;
% [illegible]:
T=1;
% [illegible] [illegible]
nh=(1:N)*2-1;
% [illegible] [illegible] [illegible] [illegible], [illegible] [illegible] cos:
Am=2/pi ./ nh;
% [illegible] [illegible]:
harmonics=cos(2 * pi * nh' * t/T);
% [illegible] [illegible] [illegible]:
s1=harmonics.*repmat(Am',1,length(t));
% [illegible] [illegible]:
s2=cumsum(s1);
% [illegible] [illegible]:
for k=1:N
subplot(4,2,k)
plot(t, s2(k,:))
end
print("meandr_sin.png")
```

(. **fig:003**).



2.3: ,

2.3 Определение спектра и параметров сигнала

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.

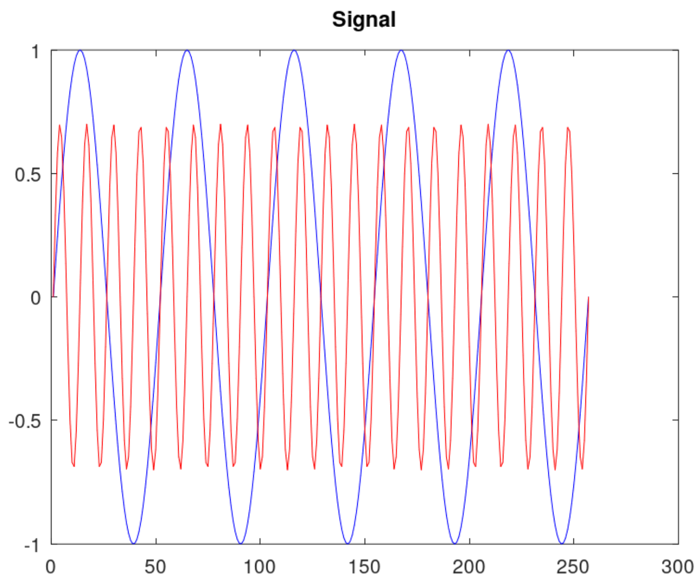
(5).

5.

```
% spectre1/spectre.m
% ⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠ signal ⊠ spectre ⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠:
mkdir 'signal';
mkdir 'spectre';
% ⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠ (⊠):
tmax = 0.5;
% ⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠⊠) (⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠):
```

```
fd = 512;
% [illegible] [illegible] [illegible] ([illegible]):
f1 = 10;
% [illegible] [illegible] [illegible] ([illegible]):
f2 = 40;
% [illegible] [illegible] [illegible]:
a1 = 1;
% [illegible] [illegible] [illegible]:
a2 = 0.7;
% [illegible] [illegible] [illegible]:
t = 0:1./fd:tmax;
% [illegible] [illegible]:
fd2 = fd/2;
% [illegible] [illegible] [illegible] [illegible]:
signal1 = a1*sin(2*pi*t*f1);
signal2 = a2*sin(2*pi*t*f2);
% [illegible] 1-[illegible] [illegible]:
plot(signal1,'b');
% [illegible] 2-[illegible] [illegible]:
hold on
plot(signal2,'r');
hold off
title('Signal');
% [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] signal:
print 'signal/spectre.png';
```

(. **fig:004**).



2.4:

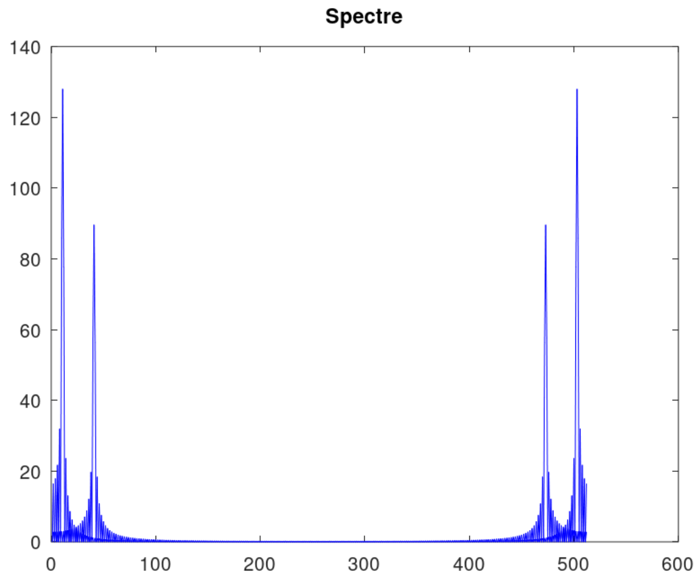
(fft()) ,

4 (5):

5.

```
%
% 1:
spectre1 = abs(fft(signal1,fd));
% 2:
spectre2 = abs(fft(signal2,fd));
% :
plot(spectre1,'b');
hold on
plot(spectre2,'r');
hold off
title('Spectre');
print 'spectre/spectre.png';
```

(. **fig:005**).



2.5:

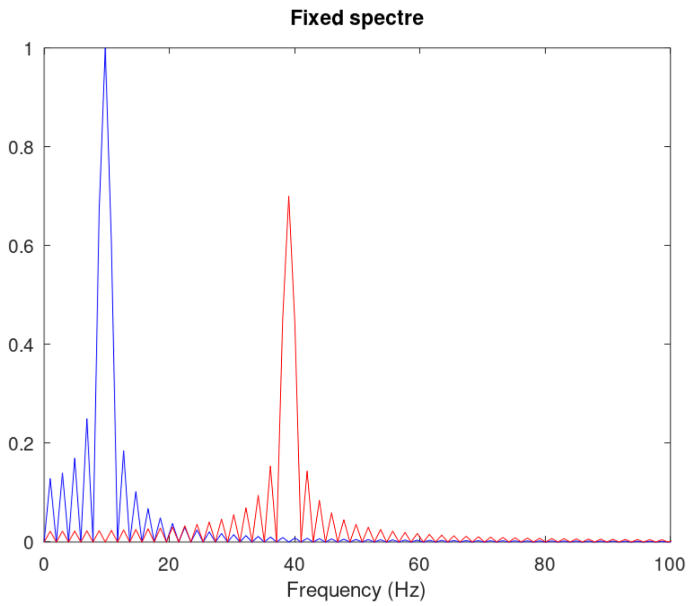
(. 5): , , . (0 250) (0 1), (6):

6.

```
% :
f = 1000*(0:fd2)./(2*fd);
% :
spectre1 = 2*spectre1/fd2;
spectre2 = 2*spectre2/fd2;
% :
plot(f,spectre1(1:fd2+1),'b');
hold on
plot(f,spectre2(1:fd2+1),'r');
```

```
hold off
xlim([0 100]);
title('Fixed spectre');
xlabel('Frequency (Hz)');
print 'spectre/spectre_fix.png';
```

(. **fig:006**).



2.6:

(7).

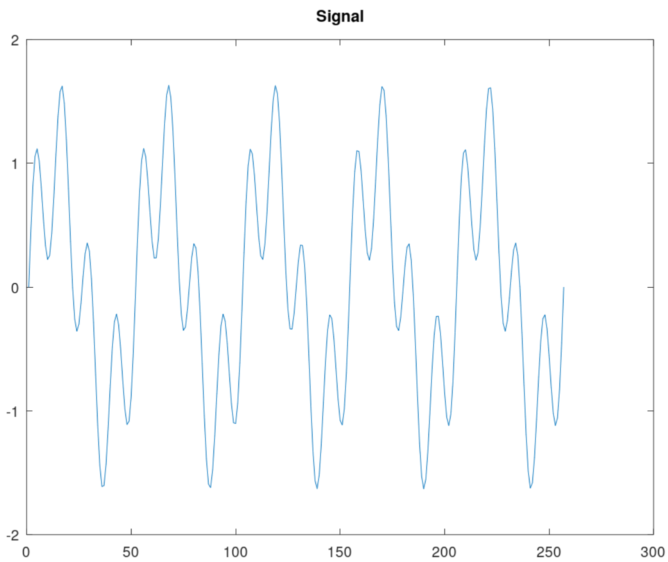
7.

```
% spectr_sum/spectre_sum.m
% ☒☒☒☒☒☒☒☒ ☒☒☒☒☒☒☒☒☒ signal ☒ spectre ☒☒☒ ☒☒☒☒☒☒☒☒☒☒ ☒☒☒☒☒☒☒☒:
mkdir 'signal';
mkdir 'spectre';
% ☒☒☒☒☒ ☒☒☒☒☒☒☒ (☒):
tmax = 0.5;
% ☒☒☒☒☒☒☒ ☒☒☒☒☒☒☒☒☒☒☒☒☒ (☒☒) (☒☒☒☒☒☒☒☒☒☒ ☒☒☒☒☒☒☒☒):
```

```
fd = 512;
% [illegible] [illegible] [illegible] ([illegible]):
f1 = 10;
% [illegible] [illegible] [illegible] ([illegible]):
f2 = 40;
% [illegible] [illegible] [illegible]:
a1 = 1;
% [illegible] [illegible] [illegible]:
a2 = 0.7;
% [illegible] [illegible]
fd2 = fd/2;
% [illegible] [illegible] [illegible] ([illegible]) [illegible] [illegible]:
% [illegible] [illegible] [illegible]:
t = 0:1./fd:tmax;
signal1 = a1*sin(2*pi*t*f1);
signal2 = a2*sin(2*pi*t*f2);
signal = signal1 + signal2;
plot(signal);
title('Signal');
print 'signal/spectre_sum.png';
% [illegible] [illegible]:
% [illegible] [illegible] [illegible] [illegible]:
spectre = fft(signal,fd);
% [illegible] [illegible]
f = 1000*(0:fd2)./(2*fd);
% [illegible] [illegible] [illegible] [illegible]:
spectre = 2*sqrt(spectre.*conj(spectre))./fd2;
% [illegible] [illegible] [illegible] [illegible]:
plot(f,spectre(1:fd2+1))
```

```
xlim([0 100]);
title('Spectre');
xlabel('Frequency (Hz)');
print 'spectre/spectre_sum.png';
```

(. **fig:007**):



2.7:

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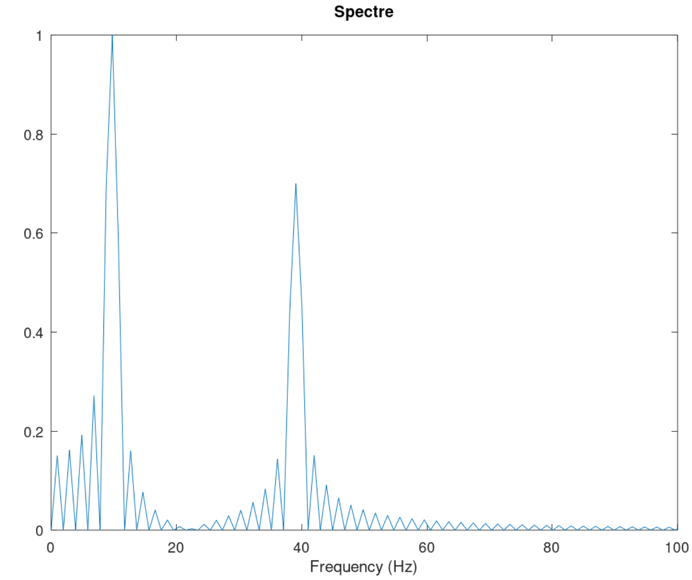
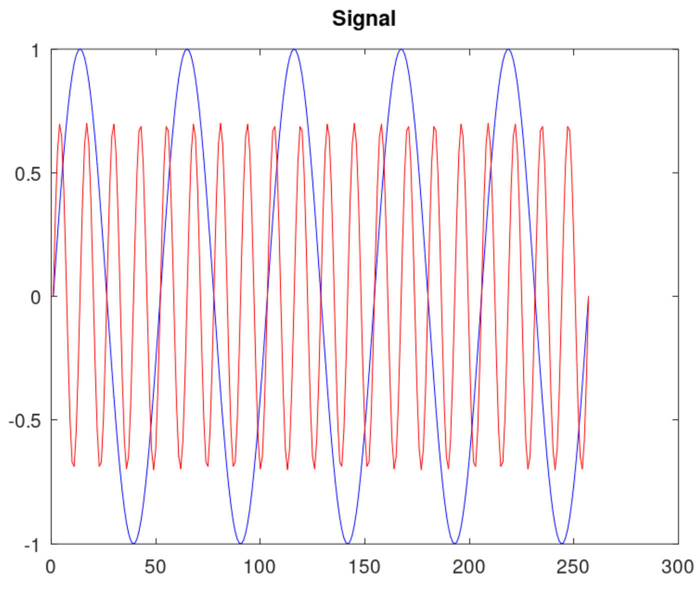
, (. **fig:008**):



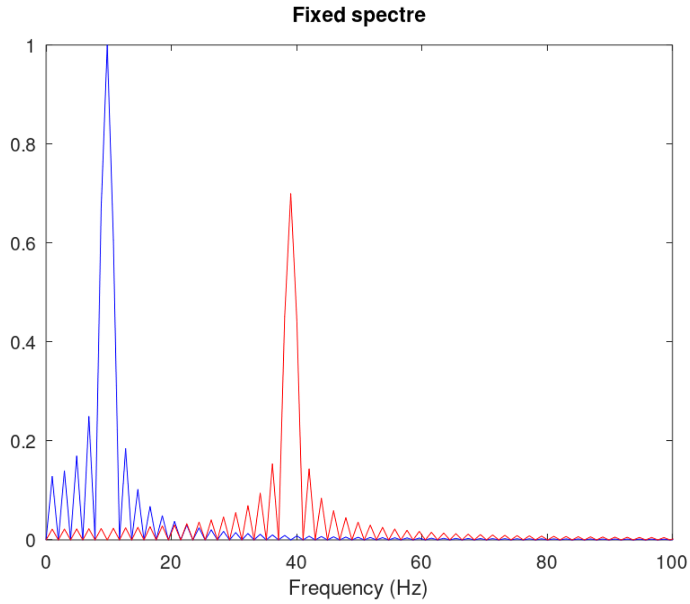
2.8:

, 5 7 fd 128.

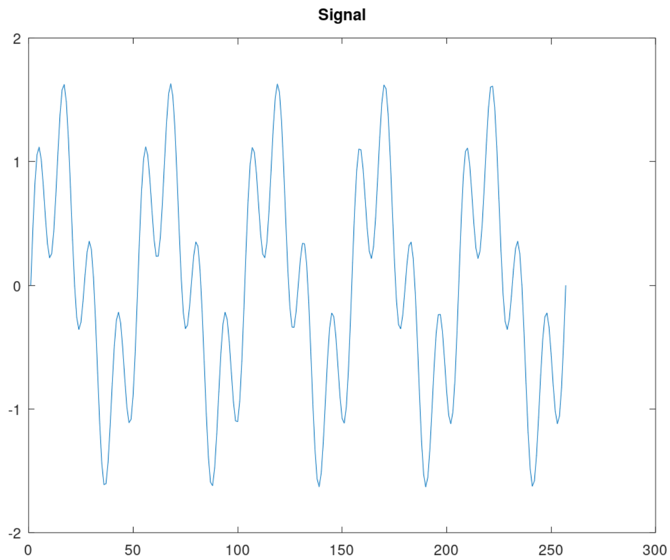
(. **fig:009**[**fig:012**]):



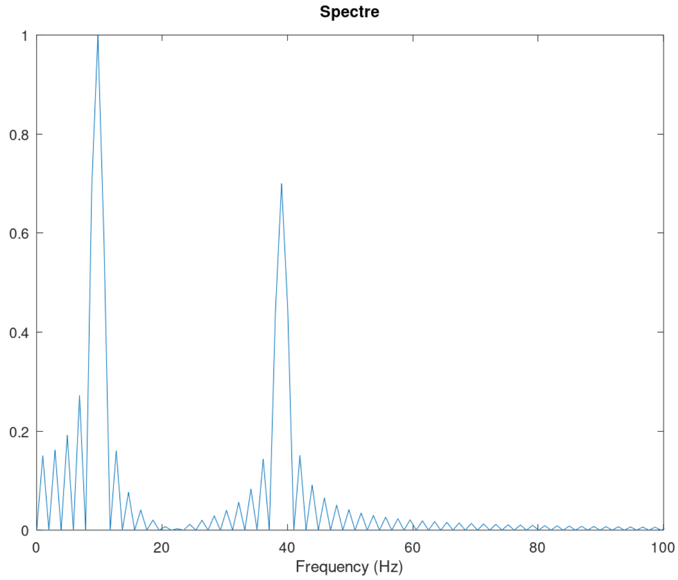
2.9: fd = 128.



2.10: fd = 128.



2.11: fd = 128.



2.12: fd = 128.

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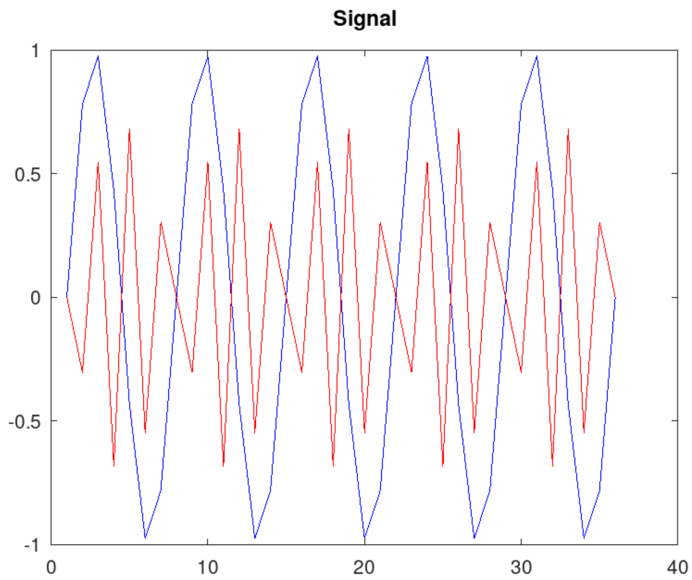
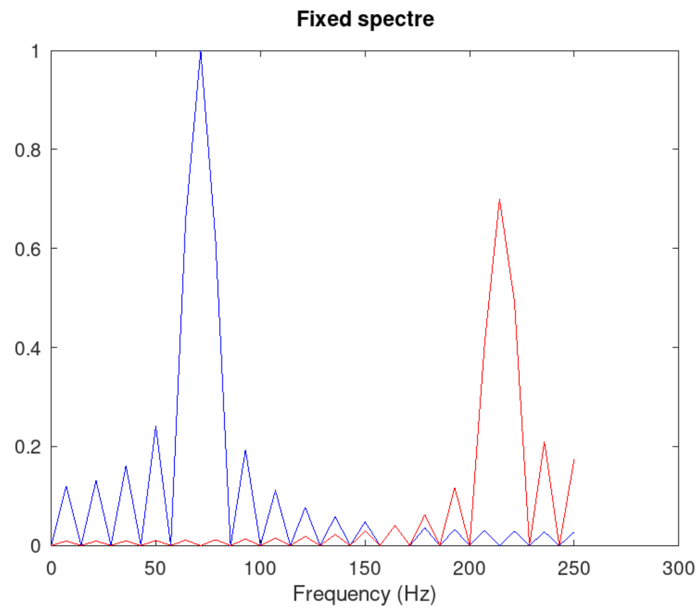
, , 80 . 5 7

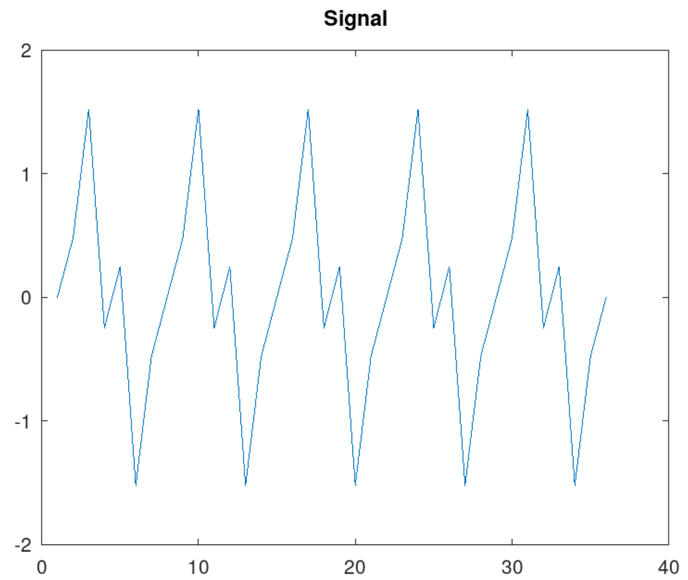
fd 70. (. **fig:013**[**fig:016**]):



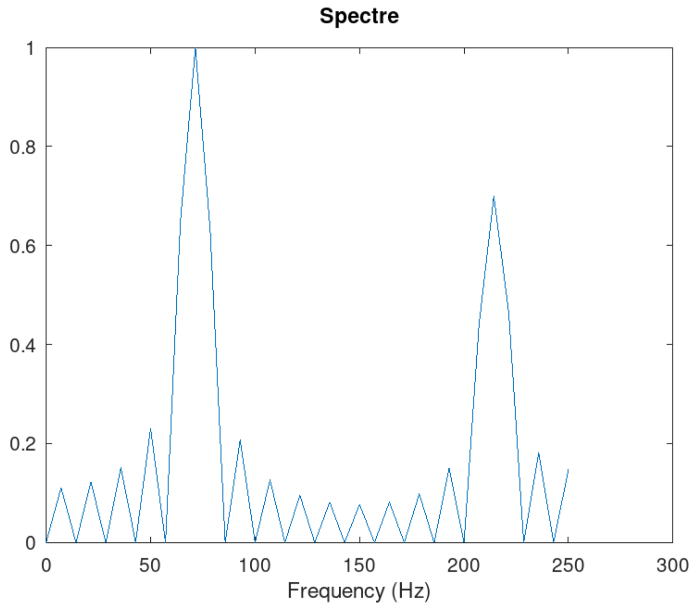
2.13: fd = 70.



2.14: fd = 70.



2.15: fd = 70.



2.16: fd = 70.

80 , ,

. 40 .

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10, .

2.4 Амплитудная модуляция

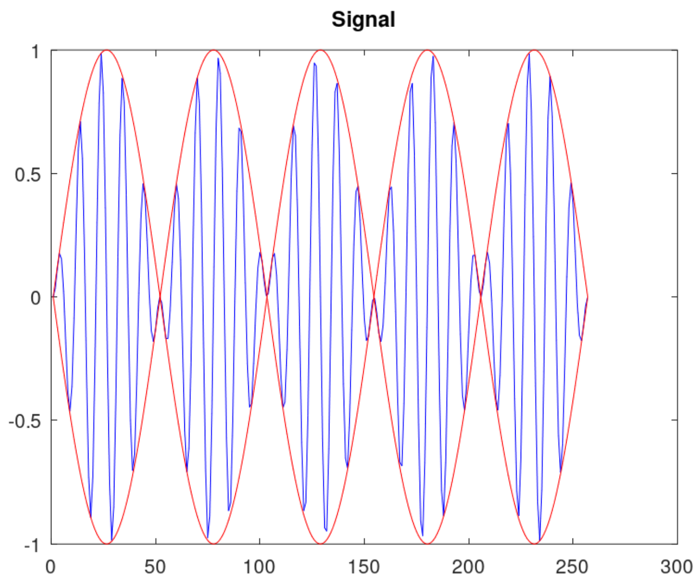
(8).

8.

```
% modulation/am.m
% [illegible] [illegible] signal [illegible] spectre [illegible] [illegible] [illegible]:
mkdir 'signal';
mkdir 'spectre';
% [illegible] [illegible] [illegible] [illegible] 50 [illegible] 5
% [illegible] [illegible] ([illegible])
tmax = 0.5;
% [illegible] [illegible] ([illegible]) ([illegible] [illegible])
fd = 512;
% [illegible] [illegible] ([illegible])
f1 = 5;
% [illegible] [illegible] ([illegible])
f2 = 50;
% [illegible] [illegible]
fd2 = fd/2;
% [illegible] [illegible] [illegible] [illegible] ([illegible])
% [illegible] [illegible]
% [illegible] [illegible] [illegible]:
t = 0:1./fd:tmax;
signal1 = sin(2*pi*t*f1);
signal2 = sin(2*pi*t*f2);
signal = signal1 .* signal2;
```

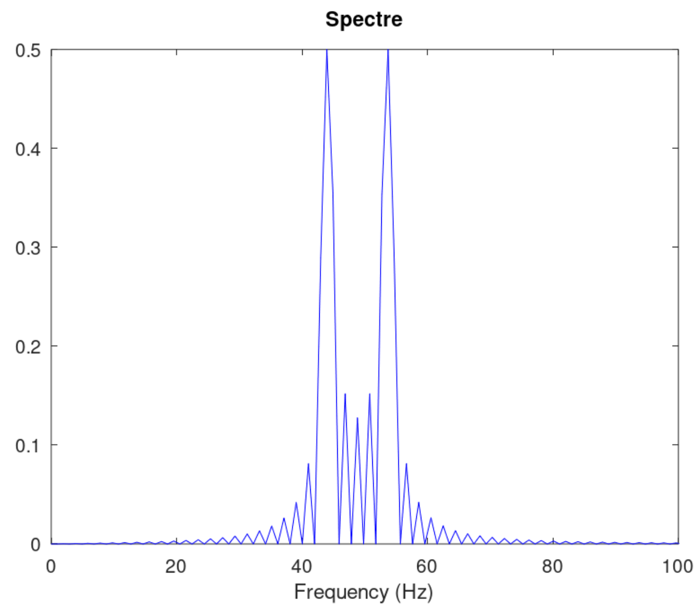
```
plot(signal, 'b');
hold on
% [illegible] [illegible]:
plot(signal1, 'r');
plot(-signal1, 'r');
hold off
title('Signal');
print 'signal/am.png';
% [illegible] [illegible]:
% [illegible] [illegible] [illegible]-[illegible]:
spectre = fft(signal,fd);
% [illegible] [illegible]:
f = 1000*(0:fd2)./(2*fd);
% [illegible] [illegible] [illegible] [illegible]:
spectre = 2*sqrt(spectre.*conj(spectre))./fd2;
% [illegible] [illegible]:
plot(f,spectre(1:fd2+1), 'b')
xlim([0 100]);
title('Spectre');
xlabel('Frequency (Hz)');
print 'spectre/am.png';
```

(. **fig:017**):



2.17:

, (. **fig:018**):



2.18:

2.5 Кодирование сигнала. Исследование свойства самосинхронизации сигнала

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coding main.m, maptowave.m, unipolar.m, ami.m, bipolarnrz.m, bipolarrz.m, manchester.m, diffmanc.m, calcspectre.m. signal.

maim.m signal (9):

9.

```
% coding/main.m
% [illegible] [illegible] signal:
pkg load signal;
% [illegible] [illegible] [illegible]:
data=[0 1 0 0 1 1 0 0 0 1 1 0];
% [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]:
data_sync=[0 0 0 0 0 0 0 1 1 1 1 1 1 1];
% [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]:
data_spectre=[0 1 0 1 0 1 0 1 0 1 0 1 0 1];
% [illegible] [illegible] signal, sync [illegible] spectre [illegible] [illegible] [illegible]:
mkdir 'signal';
mkdir 'sync';
mkdir 'spectre';
axis("auto");
```

data(10):

10. **data**

```
% [illegible] [illegible]
wave=unipolar(data);
plot(wave);
ylim([-1 6]);
title('Unipolar');
print 'signal/unipolar.png';

% [illegible] ami
wave=ami(data);
plot(wave)
title('AMI');
print 'signal/ami.png';

% [illegible] NRZ
wave=bipolarnrz(data);
plot(wave);
title('Bipolar Non-Return to Zero');
print 'signal/bipolarnrz.png';

% [illegible] RZ
wave=bipolarrz(data);
plot(wave)
title('Bipolar Return to Zero');
print 'signal/bipolarrz.png';

% [illegible] [illegible]
wave=manchester(data);
plot(wave)
title('Manchester');
```

```
print 'signal/manchester.png';

%
wave=diffmanc(data);
plot(wave)
title('Differential Manchester');
print 'signal/diffmanc.png';
```

data_sync(11):

11.

data_sync

```
%
wave=unipolar(data_sync);
plot(wave);
ylim([-1 6]);
title('Unipolar');
print 'sync/unipolar.png';

% AMI
wave=ami(data_sync);
plot(wave)
title('AMI');
print 'sync/ami.png';
% NRZ
wave=bipolarnrz(data_sync);
plot(wave);
title('Bipolar Non-Return to Zero');
print 'sync/bipolarnrz.png';
```

```
% [illegible] RZ
wave=bipolarrz(data_sync);
plot(wave)
title('Bipolar Return to Zero');
print 'sync/bipolarrz.png';
% [illegible] [illegible]
wave=manchester(data_sync);
plot(wave)
title('Manchester');
print 'sync/manchester.png';

% [illegible] [illegible] [illegible]
wave=diffmanc(data_sync);
plot(wave)
title('Differential Manchester');
print 'sync/diffmanc.png';
```

(12):

12.

```
% [illegible] [illegible]:
wave=unipolar(data_spectre);
spectre=calcspectre(wave);
title('Unipolar');
print 'spectre/unipolar.png';

% [illegible] AMI:
wave=ami(data_spectre);
spectre=calcspectre(wave);
```

```
title('AMI');
print 'spectre/ami.png';

% NRZ:
wave=bipolarnrz(data_spectre);
spectre=calcspectre(wave);
title('Bipolar Non-Return to Zero');
print 'spectre/bipolarnrz.png';

% RZ:
wave=bipolarrz(data_spectre);
spectre=calcspectre(wave);
title('Bipolar Return to Zero');
print 'spectre/bipolarrz.png';

% :
wave=manchester(data_spectre);
spectre=calcspectre(wave);
title('Manchester');
print 'spectre/manchester.png';

% :
wave=diffmanc(data_spectre);
spectre=calcspectre(wave);
title('Differential Manchester');
print 'spectre/diffmanc.png';
```

maptowave.m , (13):

13. ,

```
% coding/maptowave.m
function wave=maptowave(data)
data=upsample(data,100);
wave=filter(5*ones(1,100),1,data);
```

unipolar.m, ami.m, bipolarnrz.m, bipolarrz.m, manchester.m, diffmanc.m data maptowave (14-19).

14.

```
% coding/unipolar.m
% [illegible] [illegible]:
function wave=unipolar(data)
wave=maptowave(data);
```

15. AMI

```
% coding/ami.m
% [illegible] AMI:
function wave=ami(data)
am=mod(1:length(data(data==1)),2);
am(am==0)=-1;
data(data==1)=am;
wave=maptowave(data);
```

16. NRZ

```
% coding/bipolarnrz.m
% [illegible] NRZ:
function wave=bipolarnrz(data)
data(data==0)=-1;
wave=maptowave(data);
```

17. RZ

```
% coding/bipolarrz.m
% RZ:
function wave=bipolarrz(data)
data(data==0)=-1;
data=upsample(data,2);
wave=maptowave(data);
```

18.

```
% coding/manchester.m
% :
function wave=manchester(data)
data(data==0)=-1;
data=upsample(data,2);
data=filter([-1 1],1,data);
wave=maptowave(data);
```

19.

```
% coding/diffmanc.m
%
function wave=diffmanc(data)
data=filter(1,[1 1],data);
data=mod(data,2);
wave=manchester(data);
```

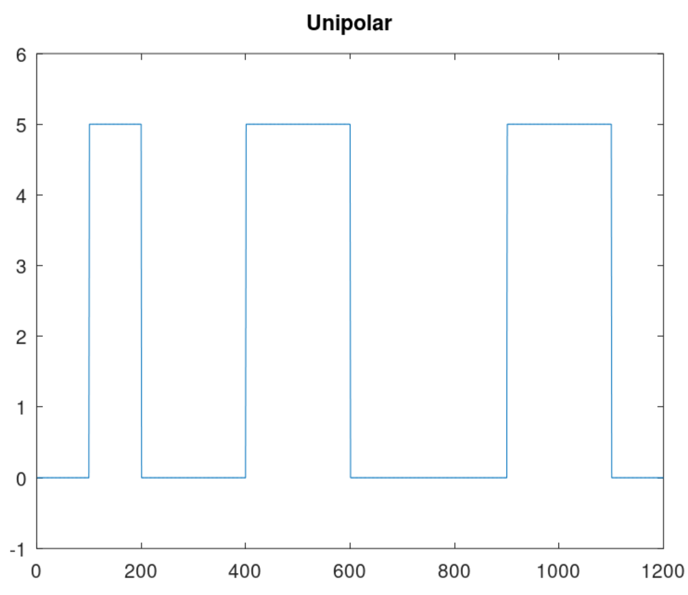
calcspectre.m (20):

20.

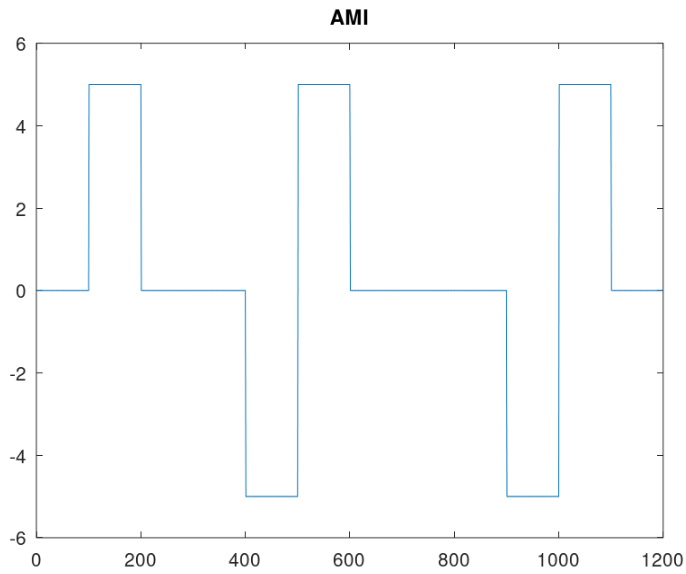
```
% calcspectre.m
% ☒☒☒☒☒☒ ☒☒☒☒☒☒☒☒☒ ☒☒☒☒☒☒ ☒☒☒☒☒☒:
function spectre = calcspectre(wave)
% ☒☒☒☒☒☒ ☒☒☒☒☒☒☒☒☒☒☒☒ (☒☒):
Fd = 512;
Fd2 = Fd/2;
Fd3 = Fd/2 + 1;

X = fft(wave,Fd);
spectre = X.*conj(X)/Fd;
f = 1000*(0:Fd2)/Fd;
plot(f,spectre(1:Fd3));
xlabel('Frequency (Hz)');
```

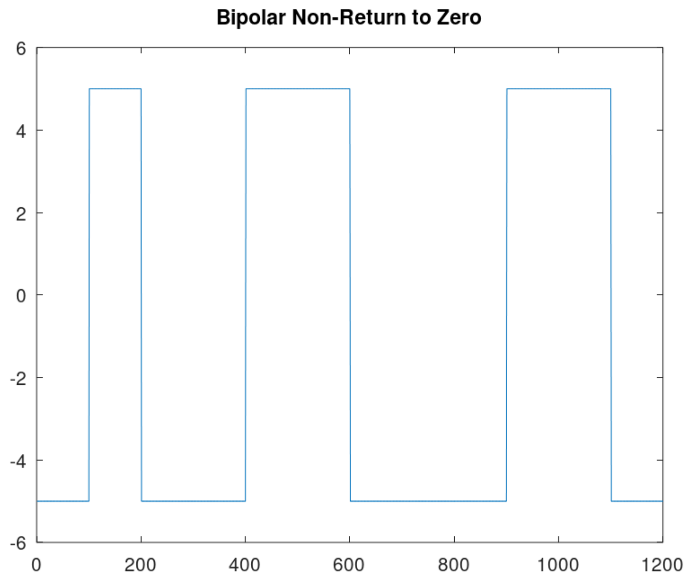
main.m. signal (. **fig:019**[**fig:024**]), sync - , (. **fig:025**[**fig:030**]), spectre — (. **fig:031**[**fig:036**]).



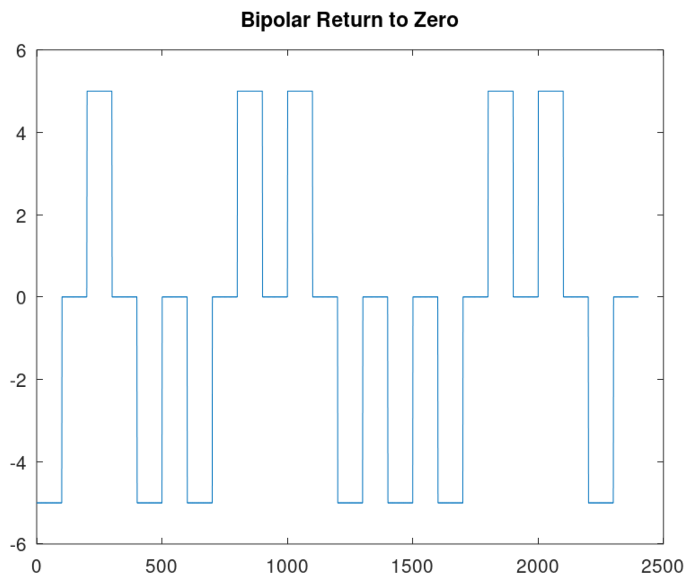
2.19:



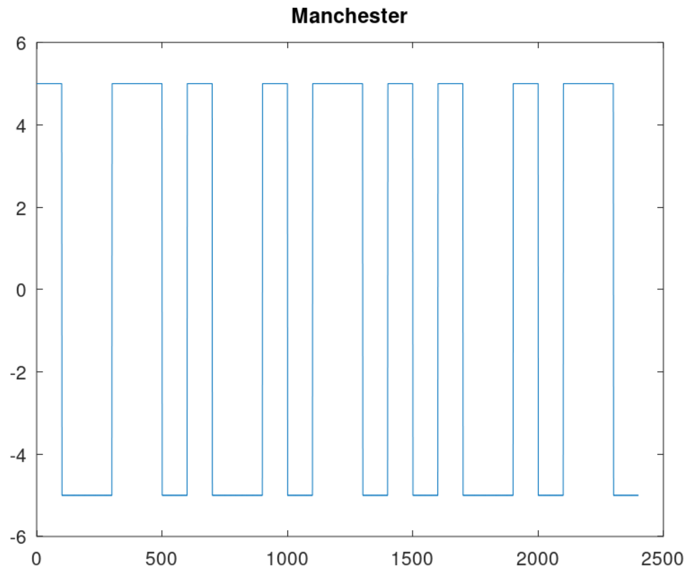
2.20: AMI



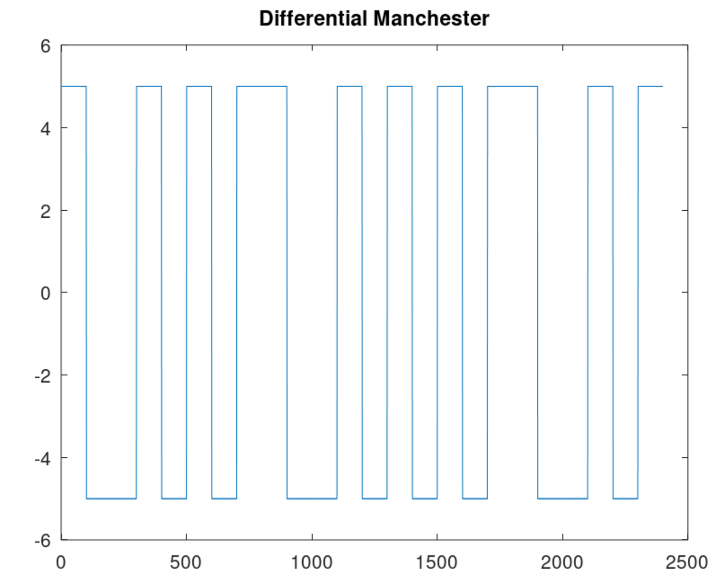
2.21: NRZ


Bipolar Return to Zero
6
4
2
0
-2
-4
-6
0
500
1000
1500
2000
2500

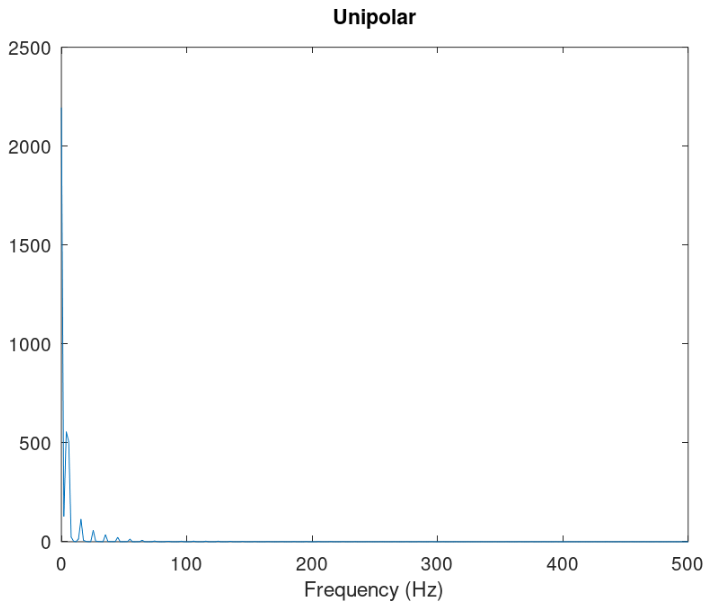
2.22: RZ


Manchester
6
4
2
0
-2
-4
-6
0
500
1000
1500
2000
2500

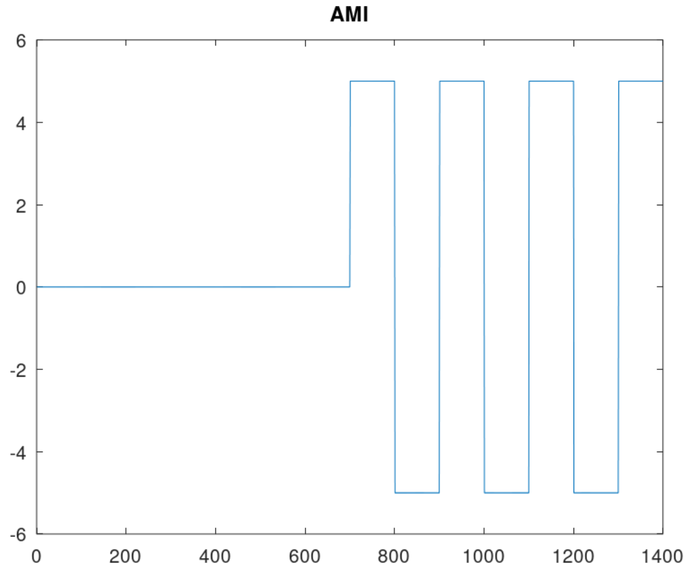
2.23:


Differential Manchester
6
4
2
0
-2
-4
-6
0
500
1000
1500
2000
2500

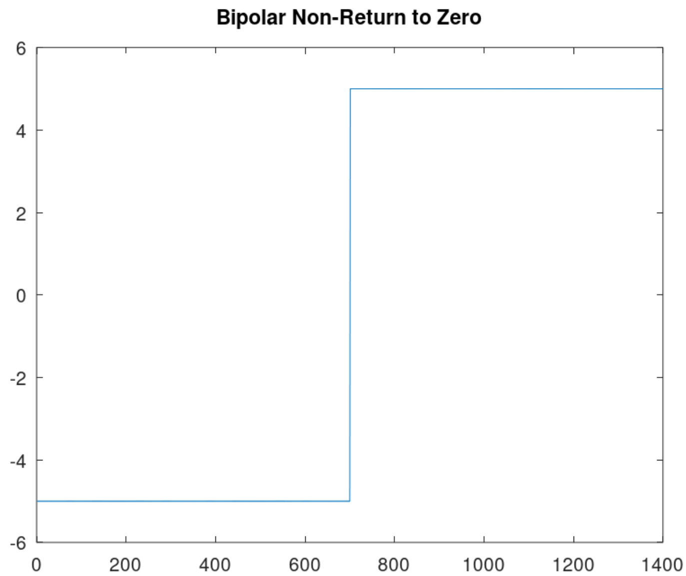
2.24:


Unipolar
2500
2000
1500
1000
500
0
0
100
200
300
400
500
Frequency (Hz)

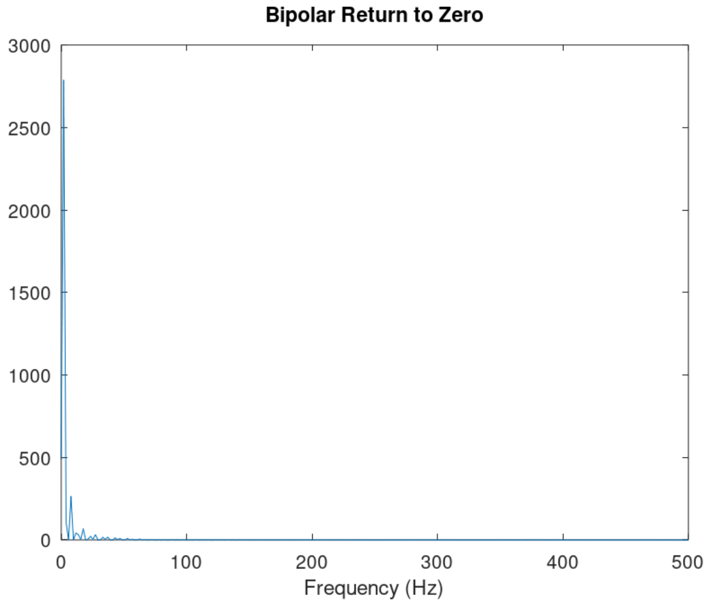
2.25: :



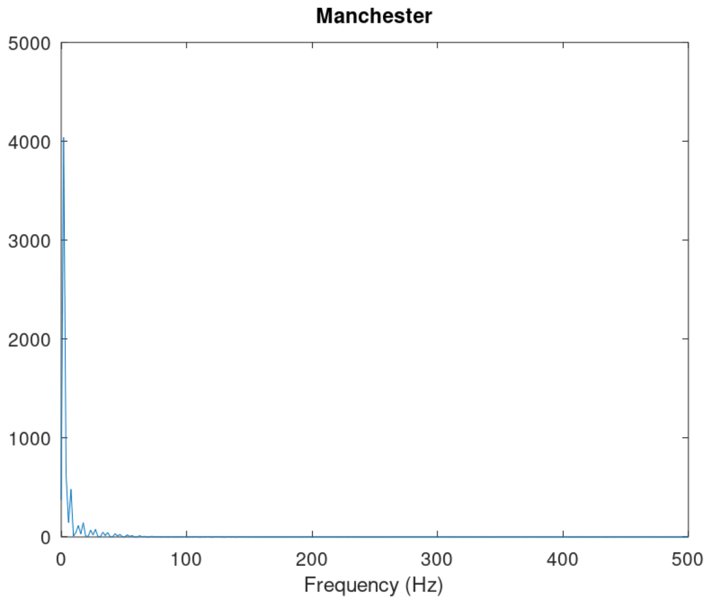
2.26: AMI:



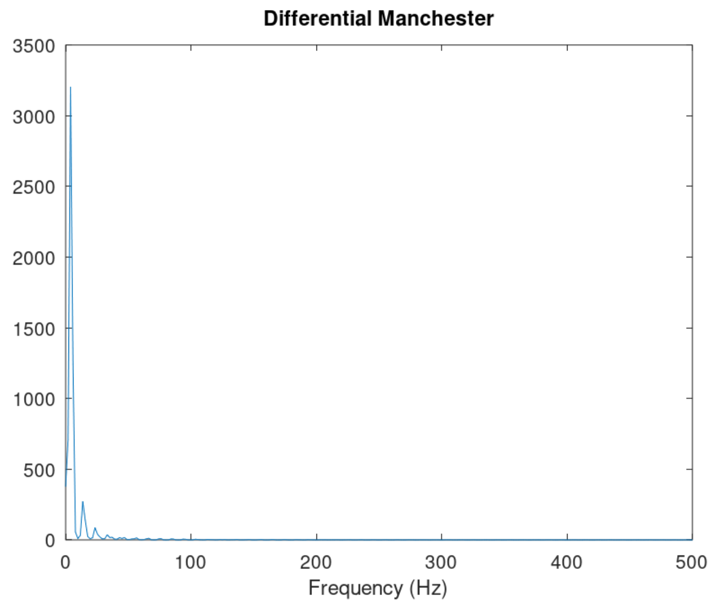
2.27: NRZ:


Bipolar Return to Zero
3000
2500
2000
1500
1000
500
0
0
100
200
300
400
500
Frequency (Hz)

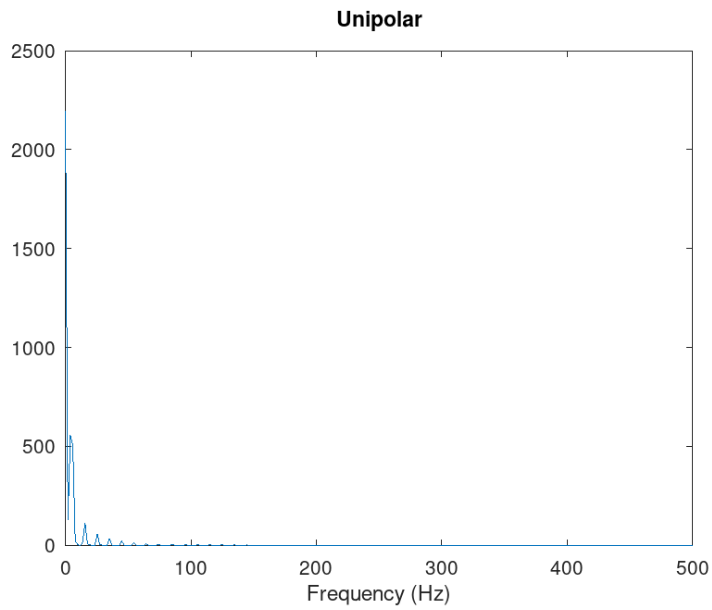
2.28: RZ:


Manchester
5000
4000
3000
2000
1000
0
0
100
200
300
400
500
Frequency (Hz)

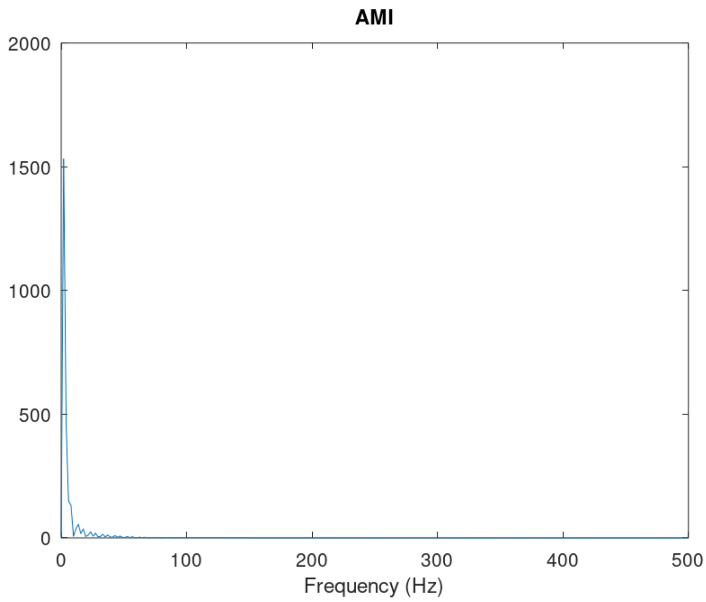
2.29: :


Differential Manchester
3500
3000
2500
2000
1500
1000
500
0
0
100
200
300
400
500
Frequency (Hz)

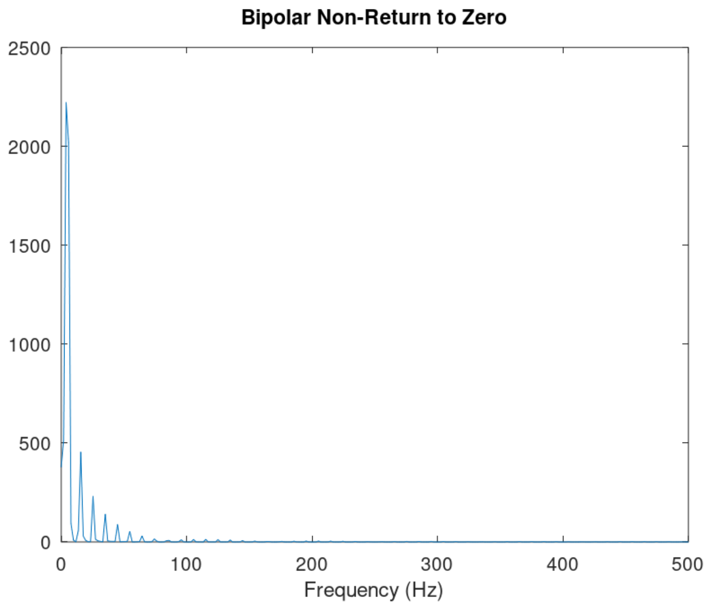
2.30: :


Unipolar
2500
2000
1500
1000
500
0
0
100
200
300
400
500
Frequency (Hz)

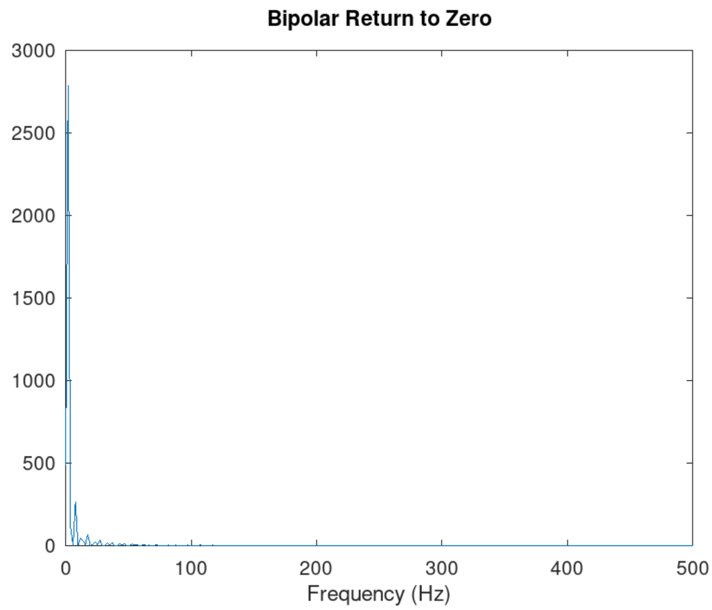
2.31: :



2.32: AMI:

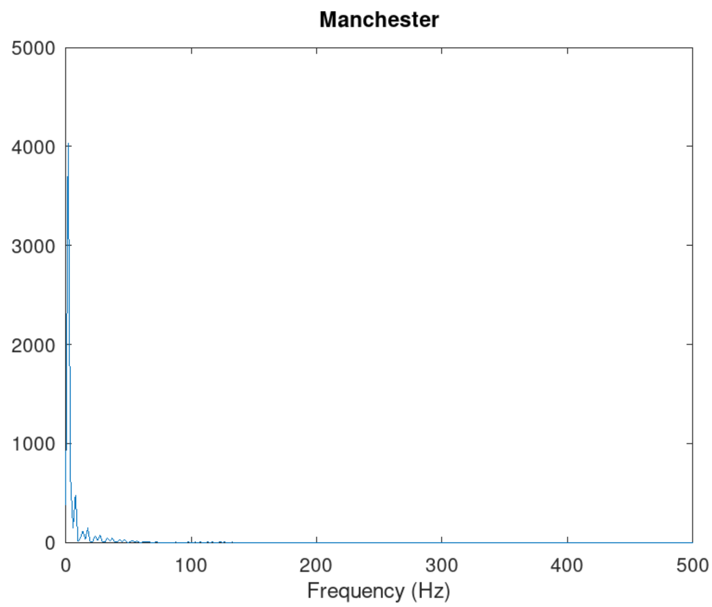


2.33: NRZ:



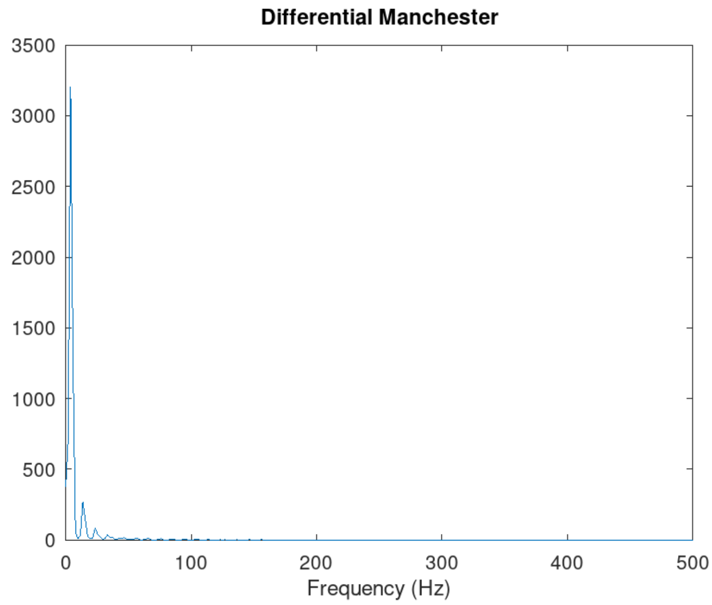
Bipolar Return to Zero
3000
2500
2000
1500
1000
500
0
0
100
200
300
400
500
Frequency (Hz)

2.34: RZ:



Manchester
5000
4000
3000
2000
1000
0
0
100
200
300
400
500
Frequency (Hz)

2.35: :


Differential Manchester
3500
3000
2500
2000
1500
1000
500
0
0
100
200
300
400
500
Frequency (Hz)

2.36: :

3 Выводы

- Octave
- ,

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